

Toward a Phonetic Decryption of the Voynich Manuscript: A Computational Pipeline Using King-Andrisani Transliteration and Agglutinative Segmentation

Guillaume Clement
Flow Line Integration | gcle1979@gmail.com
Assisted by Claude (Anthropic, Opus 4.6)

Abstract

The Voynich Manuscript (Beinecke MS 408, Yale University), radiocarbon dated to 1404-1438, has resisted decipherment for over a century. This paper presents K&A v12, a computational pipeline for Latin phonetic recovery from EVA-transcribed text, building upon the King-Andrisani transliteration hypothesis (2021) which proposes that the Voynich script represents a modified subset of Tironian Notes encoding Latin phonetic values. While this foundational hypothesis remains subject to independent verification, our pipeline provides a systematic framework for testing it at scale.

The pipeline introduces three methodological innovations. First, a monolithic-first decoding strategy attempts to match entire EVA tokens against the Perseus Digital Library Latin dictionary (265,419 attested forms) before resorting to segmentation, thereby preserving short, high-frequency pharmaceutical terms such as *aquam*, *cura*, and *coquere*. Second, an agglutination-aware prefix detection system, inspired by Arabic proclitic attachment, identifies 13 prefix rules whereby Latin prepositions appear to be fused to the following word (e.g., EVA prefix *y* mapping to Latin *in*, with 654 confirmed prefix-win matches). Third, a 9-signal calibrated scorer integrates Perseus validation, corpus frequency, morphological plausibility, and collision penalties to select optimal decode candidates.

Applied to all 226 folio sides (38,442 words), the pipeline achieves 89.3% Perseus dictionary validation and 90.6% CONFIRMED+HIGH confidence grading. These results are consistent with, though not proof of, the hypothesis that the underlying text is Latin.

Structural analysis of folio f57v reveals a circular diagram whose five-layer architecture (54-element outer ring, 4x17 repeating pattern, 29-word ring corresponding to the synodic month, 75%-arc sundial, central pivot) exhibits near-perfect structural isomorphism with the Ashmole 370 Kalendarium (~1424, Bodleian Library), suggesting the page may function as a medico-astrological volvelle for timing therapeutic interventions.

Pharmacological cross-validation against the Antidotarium Nicolai (12th-century Salernitan formulary) identifies 7 of 12 canonical Aurea Alexandrina ingredients on folio f103r (532 words, 91% Perseus validation), including *aloe*, *ture* (frankincense), *sal*, *olei*, *cerae*, and the preparation verbs *coque* (cook, 17+ occurrences) and *cola* (strain). Systematic folio-by-folio analysis reveals additional pharmaceutical vocabulary on astronomical pages: *nardi* (spikenard, f67r2), *cassiae* (cinnamon, f67v1), *apii* (celery, f85r1), and *vini* (wine, f67r1). In total, 33 pharmaceutical terms are identified across the corpus. The presence of Italian vernacular forms (*pepe* for *piper*, *lilie* for *lilium*) is consistent with a Northern Italian scribe, aligning with paleographic analysis.

We further observe evidence for a dual cipher system: a verbose phonetic substitution for body text, and

a separate nomenclator (codebook) for proper nouns and astronomical entities, as demonstrated by the systematic failure of zodiacal crib attacks on folios f70v1-f73v.

Significant limitations remain. Approximately 9% of words remain undecoded. Four Aurea Alexandrina ingredients have not been identified. The nomenclator layer is unresolved. The homophonic cipher produces collisions between distinct EVA tokens. Independent replication by domain experts in medieval Latin and pharmaceutical history is essential before any claims of decipherment can be considered established.

The complete pipeline code (Python), decoded output for all folios, and an enriched 226-folio catalogue are publicly available for independent verification.

Keywords: *Voynich Manuscript, Latin decryption, King-Andrisani transliteration, agglutination, phonetic cipher, Perseus validation, Antidotarium Nicolai, volvelle, computational philology, human-AI collaboration*

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